# ProprioVLA: Self-Supervised Proprioceptive Grounding for Vision-Language-Action Policies

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## Abstract

Vision-language-action (VLA) policies have shown strong promise as generalist controllers for robot manipulation, yet their action predictions can remain fragile under changes in visual perspective. We identify a systematic failure pattern that lateral camera shifts induce co-directional shifts in predicted robot trajectories, suggesting that current VLAs may exploit image-space shortcuts, such as pixel-coordinate correlations, rather than grounding actions in proprioceptive state and robot-object geometry. To bridge this gap, we introduce PROPRIOVLA, a self-supervised alignment approach that incentivizes self-grounding capability in VLA policies. PROPRIOVLA implicitly encourages the policy to localize its own embodiment by aligning learned grounding queries with state-aligned residual flow targets mined from robot interaction. Specifically, PROPRIOVLA turns proprioceptively aligned visual change, captured by optical flow and image difference, into token-level self-grounding targets. It then trains a lightweight learned-query head over visual tokens in the VLA representation to recover these targets with normalized KL supervision, regularizing the visual representation to encode the visible embodiment without manual masks, external grounding datasets, or stronger foundation-model supervision. Extensive experiments on both real-robot and simulation environments show that PROPRIOVLA improves task success and robustness over action-supervised VLA adaptation, suggesting that proprioceptive grounding is a practical route toward spatially reliable VLA control.

## 1 Introduction

Vision-language-action (VLA) policies make language-conditioned robot manipulation increasingly general, but reliable control requires spatial grounding in the robot's own embodiment. Recent robot foundation models show that broad pretraining and large robot datasets can yield policies that transfer across tasks and embodiments [2, 3, 7, 5]. For manipulation, however, recognizing the instructed object is not sufficient: the policy must relate visual evidence to proprioceptive state and to the geometry of the surrounding scene. Spatial robustness under changes in visual perspective is therefore a core requirement for dependable VLA control.

We make this requirement concrete with two diagnostic findings. First, when the physical robot-scene state is fixed and only the camera viewpoint is laterally shifted, VLA-predicted trajectories shift in the same image direction. Second, even robot-centric prompts such as "Where is the robot itself?" often produce attention maps that fail to localize the gripper or end-effector. Together, these findings reveal that task grounding does not imply self-grounding: a policy may recognize the instructed object while failing to bind its actions to the visible body that executes them.

This missing-self failure suggests a root technical issue in current VLA adaptation. A policy trained only to match demonstrations under a fixed observation distribution can learn image-space shortcuts,

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such as pixel-coordinate correlations, that entangle the action frame with the image frame. In that case, proprioception is present but not visually grounded: the model receives the robot state, yet its visual representation does not reliably encode where the acting body appears. Spatially reliable manipulation therefore requires a learning signal that binds proprioceptive change to visual evidence of the robot itself.

Existing grounding-oriented VLA methods show that perception, attention allocation, and embodiment cues must be treated as first-class control problems. Recent methods improve spatial awareness with visual trajectory prompts, 3D representations, camera-space action formulations, view synthesis, or explicit treatments of proprioception [11, 8, 9, 4, 10, 6]. These approaches are valuable, but they do not directly provide a self-supervised signal that tells the policy where its own acting body appears in the image. The missing cue is not only object grounding or view robustness, but self-grounding: a binding between proprioceptive state and visual evidence of the visible embodiment.

PROPRIOVLA addresses this missing cue by learning self-supervised proprioceptive grounding from robot interaction. The key insight is that the robot reveals itself through visual change synchronized with changes in proprioceptive state. Rather than using proprioception as a passive input token, PROPRIOVLA turns proprioceptively aligned visual change, captured by optical flow and image difference, into token-level self-grounding targets. These targets provide implicit supervision for where the visible embodiment appears, without manual masks, external grounding datasets, or stronger foundation-model supervision.

The method realizes this idea through two components. State-Aligned Residual Flow (SARF) mines confidence-weighted token-level self-grounding targets from robot interaction by retaining residual visual change that is both locally salient and aligned with proprioceptive change. A Proprioceptive Alignment Module (PAM) then trains a lightweight learned-query grounding head over visual tokens in the VLA representation to recover these targets with normalized KL supervision. This auxiliary objective regularizes the visual representation to encode the visible embodiment while leaving the standard action-supervised VLA objective intact.

Experiments test whether proprioceptive self-grounding improves spatial reliability beyond ordinary action supervision and state concatenation. We evaluate PROPRIOVLA in real-robot and simulation environments, including controlled camera-perturbation settings that directly probe view robustness. Across these evaluations, PROPRIOVLA improves task success and robustness over action-supervised VLA adaptation. Ablations further support the two-part design: SARF provides the self-grounding targets, and PAM uses normalized KL supervision to stabilize learned-query grounding across target scales. These results suggest that self-supervised proprioceptive grounding is a practical route toward spatially reliable VLA control.

Our contributions are:

- We identify a missing-self failure mode in VLA policies: camera-only viewpoint shifts can induce co-directional shifts in predicted robot trajectories, while robot-centric prompting can still fail to localize the gripper or end-effector.
- We introduce State-Aligned Residual Flow (SARF) as implicit supervision for self-grounding, turning proprioceptively aligned optical-flow and image-difference cues into token-level targets without manual masks, external grounding datasets, or stronger foundation-model supervision.
- We propose PROPRIOVLA, a self-supervised proprioceptive alignment approach whose Proprioceptive Alignment Module (PAM) trains a learned-query grounding head with normalized KL supervision, improving task success and robustness in real-robot and simulation environments.

## 2 Related Work

**Vision-language-action models.** Vision-language-action (VLA) policies scale language-conditioned robot control by coupling vision-language backbones with robot action prediction. RT-1 and RT-2 established the value of large robot datasets and web-scale semantic priors [2, 3]; recent open and generalist models further improve transfer across tasks, embodiments, and action parameterizations [7, 5, 1]. However, this line of work mainly optimizes task success under the training observation distribution. It does not directly test whether the action-relevant visual evidence

is anchored to camera-invariant robot-object geometry. PROPRIOVLA studies this missing failure mode: under camera perturbations, a fine-tuned VLA can follow image-coordinate shortcuts instead of grounding actions in the robot's own state.

**Visual grounding in VLA policies.** Recent work improves VLA perception by making space, viewpoint, or proprioception more explicit. TraceVLA adds visual trajectory prompts to increase spatial-temporal awareness [11]; SpatialVLA injects 3D spatial representations [8]; OC-VLA predicts actions in camera space to reduce observation-action mismatch [9]; and VistaBot uses geometry-aware view synthesis for cross-view robustness [4]. These methods show that perception is a central bottleneck, but they depend on explicit prompts, 3D encodings, camera coordinate transforms, or generative view modules. A complementary line questions proprioception itself: State-free Policy removes proprioceptive inputs to avoid trajectory overfitting [10], while GAP shows that proprioceptive gradients can suppress visual learning during motion transitions [6]. PROPRIOVLA takes a different stance: it does not discard proprioception or treat it as a passive token. Instead, it uses proprioceptive change to mine token-level self-grounding targets, turning the robot's own motion into self-supervision for learned visual attention.

# 3 Does a VLA Know Where It Is?

A manipulation policy must know where it is, not only what object the language instruction names. We probe this self-location requirement at two levels. The action-level probe asks whether a predicted trajectory remains tied to the fixed robot-scene state when only the camera view is shifted. The representation-level probe asks whether robot-relevant attention localizes the arm, gripper, or end-effector rather than only task objects or background context. Together, these probes expose a *missing-self* failure mode: a policy can appear task-grounded while lacking a stable visual grounding of the body that acts.

## 3.1 Finding 1: The Action Frame Is Entangled with the Image Frame

We diagnose action-frame grounding by perturbing the view while holding the physical manipulation state fixed. Given observation $o_t$, instruction $x$, proprioceptive state $s_t$, and policy $\pi_\theta$, let $\tau_\theta = \pi_\theta(o_t, x, s_t)$ be the predicted trajectory from the original view. We then apply a lateral camera or image shift $T_{\Delta v_{\mathrm{cam}}}$ and compute $\tau'_\theta = \pi_\theta(T_{\Delta v_{\mathrm{cam}}}(o_t), x, s_t)$. The instruction, robot state, object configuration, and physical scene are unchanged; only the image frame moves.

If the action frame were grounded in robot-scene geometry, the prediction would remain stable under this camera-only perturbation, up to the known transform used for visualization. A missing-self failure appears when the predicted end-effector displacement follows the image displacement. We quantify this effect after projecting the trajectory change onto a common lateral coordinate system:

$$\mathrm{TSR} = \frac{\|\Delta\tau_{\mathrm{pred}}\|_2}{\|\Delta v_{\mathrm{cam}}\|_2 + \epsilon},$$
$$\mathrm{SDA} = \frac{\langle \Delta\tau_{\mathrm{pred}}, \Delta v_{\mathrm{cam}} \rangle}{\|\Delta\tau_{\mathrm{pred}}\|_2 \|\Delta v_{\mathrm{cam}}\|_2 + \epsilon}.$$

High TSR means the prediction changes strongly under a view-only shift; high SDA means the change is co-directional with the image shift. The combination indicates that the learned action frame is partially entangled with the image frame rather than fully anchored to proprioceptive state and robot-object geometry.

## 3.2 Finding 2: Task Grounding Does Not Imply Self-Grounding

We next test whether action-relevant representations localize the acting body. For a VLA or its vision-language backbone, we use robot-centric prompts such as "Where is the robot arm?", "Please identify the robot itself in the scene," and "Where is the manipulator?" We then visualize attention heatmaps, attention rollout, Grad-CAM maps, cross-attention maps, or token relevance maps. The same analysis can be applied to policy tokens used for action prediction, comparing task relevance with self relevance.

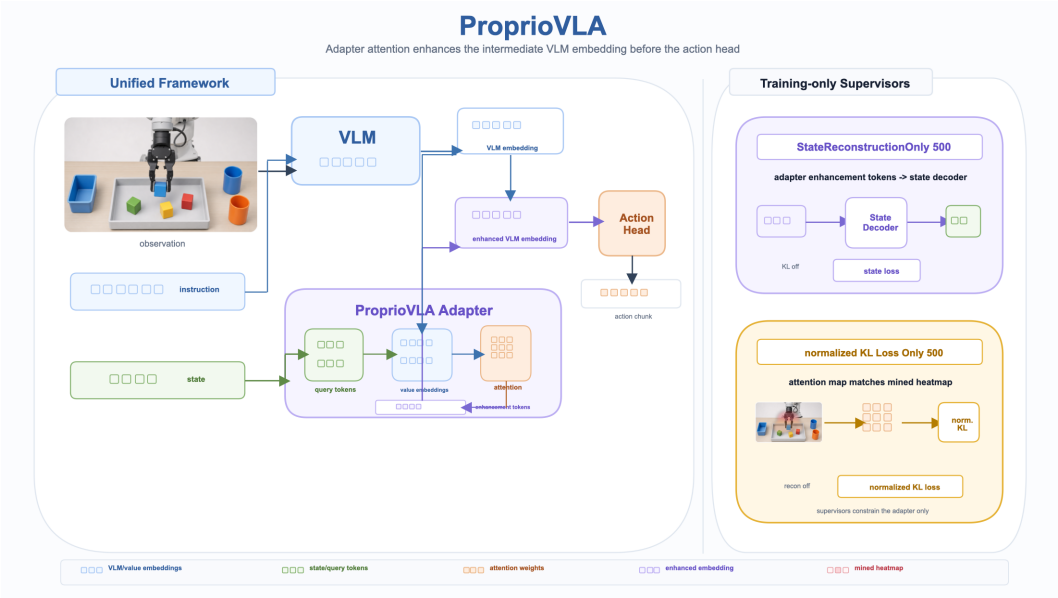


Figure 1: Overview of PROPRIOVLA. Given an observation, instruction, and proprioceptive state, the base VLA produces a standard multimodal representation. State-Aligned Residual Flow (SARF) mines token-level self-grounding targets from proprioceptively aligned visual change, while the Proprioceptive Alignment Module (PAM) trains learned grounding queries over VLA visual tokens to recover these targets. The mined target distribution is used only during training.

This probe separates task grounding from self-grounding. A model may attend to task objects, the tabletop, or salient background regions while assigning little relevance to the robot arm, gripper, or end-effector. Such attention can support object recognition and instruction following, but it does not bind the action to the body that will execute it. Robust VLA control therefore requires binding proprioceptive state to visual evidence of the acting body, rather than treating proprioception as a passive input that the visual pathway can ignore.

# 4 PROPRIOVLA

PROPRIOVLA adds an auxiliary proprioceptive grounding objective to a VLA policy while leaving the standard action-supervised objective intact. Given an observation $o_t$, language instruction $x$, and proprioceptive state $s_t$, the base VLA produces a multimodal representation containing a subset of visual tokens. PROPRIOVLA uses robot interaction to mine token-level self-grounding targets over these visual tokens and trains a lightweight learned-query grounding head to recover them. The mined targets are used only during training.

## 4.1 State-Aligned Residual Flow

State-Aligned Residual Flow (SARF) turns proprioceptively aligned visual change into self-grounding supervision. The robot reveals itself through image changes that are synchronized with changes in proprioceptive state. Given consecutive observations $(o_t, o_{t+1})$ and proprioceptive states $(s_t, s_{t+1})$, we define $\Delta s_t = s_{t+1} - s_t$. We estimate dense optical flow $F_t(p)$ between frames and subtract global or background motion to obtain residual flow $R_t(p)$ at image location $p$. We also compute a temporal image-difference cue $D_t(p)$ from the same frame pair. Together, residual flow and image difference identify where visual content changes, while proprioceptive alignment distinguishes changes likely caused by the robot itself.

SARF converts these cues into a confidence-weighted target distribution over VLA visual tokens. Let $\mathrm{sal}(D_t(p), R_t(p))$ denote local visual-change saliency from temporal image difference and residual flow. We measure alignment between residual visual motion and proprioceptive change as

$$\alpha_t(p) = \left[\cos\big(\psi_R(R_t(p)), \psi_s(\Delta s_t)\big)\right]_+ ,$$

where $\psi_R$ and $\psi_s$ are lightweight projections into a shared alignment space. The unnormalized self-grounding score is

$$e_t(p) = \mathrm{sal}(D_t(p), R_t(p))\, \alpha_t(p).$$

After aggregating $e_t$ to the VLA visual-token grid, we normalize over valid visual tokens $\mathcal{P}_t$ to obtain the target distribution

$$H_t(p) = \frac{e_t(p)}{\sum_{q \in \mathcal{P}_t} e_t(q) + \epsilon}, \qquad p \in \mathcal{P}_t.$$

We also compute a confidence $c_t \in [0, 1]$ from the total aligned visual-change mass. Low confidence downweights frames where the self-motion cue is weak or ambiguous. Thus the target is mined from robot interaction alone, without manual masks, external grounding datasets, or stronger foundation-model supervision.

### 4.2 Proprioceptive Alignment Module

The Proprioceptive Alignment Module (PAM) is a lightweight learned-query grounding head over visual tokens in the VLA representation. Let $V_t = \{v_t(p)\}_{p \in \mathcal{P}_t}$ be the valid visual tokens produced by the VLA backbone, and let $Q = \{q_m\}_{m=1}^{M}$ be $M$ learned grounding queries. PAM projects the queries and visual tokens into a shared space and uses cross-attention to predict where the visible embodiment appears:

$$A_t(m, p) = \mathrm{softmax}_{p \in \mathcal{P}_t}\left(\frac{\bar{q}_m \bar{k}_t(p)^\top}{\tau}\right), \qquad A_t(p) = \frac{1}{M}\sum_{m=1}^{M} A_t(m, p),$$

where $\bar{q}_m$ and $\bar{k}_t(p)$ are $\ell_2$-normalized query and visual-token projections, and $\tau$ is a temperature. The averaged distribution $A_t$ is the model's predicted self-grounding distribution over visual tokens.

SARF supervises PAM with a normalized KL grounding loss. Let $U_t$ denote the uniform distribution over valid visual tokens. The auxiliary grounding loss is

$$\mathcal{L}_{\text{ground}} = c_t \frac{\mathrm{KL}(H_t \,\|\, A_t)}{\mathrm{KL}(H_t \,\|\, U_t) + \epsilon}.$$

The denominator makes the loss comparable across token counts and target sharpness: matching the target approaches zero, while uniform attention is normalized to approximately one. The complete training objective is

$$\mathcal{L} = \mathcal{L}_{\text{act}} + \lambda_g \mathcal{L}_{\text{ground}},$$

where $\mathcal{L}_{\text{act}}$ is the standard imitation loss used to fine-tune the VLA. This objective regularizes the visual representation to encode the visible embodiment, while the pseudo-targets and grounding loss are not required at inference time.

## 5 Experiments

The experiments test whether proprioceptive self-grounding improves spatial reliability beyond ordinary action supervision and state concatenation. The evaluation links three evidence types: task performance, action stability under camera perturbations, and visual grounding of the acting robot.

### 5.1 Experimental Setup

We evaluate a VLA backbone fine-tuned on language-conditioned manipulation demonstrations in simulation and on a real robot. Each episode contains visual observations $o_t$, proprioceptive state $s_t$, language instruction $x$, and action supervision $a_t$. For robustness evaluation, we keep the physical robot-scene state, instruction, and proprioceptive state fixed, then apply lateral view shifts of controlled magnitude. This isolates camera-induced changes from changes in the underlying manipulation state.

We compare four policies under the same demonstrations, action labels, and view-shift protocol: Base VLA, action-supervised fine-tuning, VLA with proprioception concatenation, and PROPRIOVLA. We report task success rate, action prediction error, Trajectory Shift Ratio, Shift Direction Agreement, and

a grounding score. The grounding score is computed as either attention-target alignment with $H_t$ or attention mass on robot regions. Lower trajectory shift and direction agreement indicate a more stable action frame; higher grounding score indicates stronger localization of the acting body. Each setup reports the task suite, demonstration count, train/test split, camera shifts, action parameterization, proprioceptive normalization, optimizer, and the grounding loss weight $\lambda_g$.

### 5.2 Simulation Experiments

Simulation provides controlled access to camera shifts, object layouts, and ground-truth robot geometry. We evaluate normal, mild-shift, medium-shift, and large-shift settings. For each setting, we report task success, action prediction error, Trajectory Shift Ratio, Shift Direction Agreement, and grounding score. Robustness curves plot these metrics against shift magnitude, making the co-movement between predicted actions and image shifts directly visible.

We also evaluate held-out object layouts and camera positions. These splits separate memorization of a fixed training camera from grounding that transfers across views. Qualitative trajectory overlays compare the original prediction with shifted-view predictions after mapping them to the same robot-centric coordinate frame.

### 5.3 Real-World Experiments

The real-robot evaluation tests whether the same failure mode appears under sensor noise, calibration error, and real scene variation. We evaluate several manipulation tasks under the original view and shifted views, reporting task success, success drop under perturbation, action prediction error on logged demonstrations, and grounding score.

For qualitative analysis, we visualize baseline attention, the mined state-aligned target distribution, PROPRIOVLA attention, and trajectory overlays for the same scene. The comparison checks whether attention remains dominated by task objects and background context or shifts toward the arm, gripper, and end-effector while the trajectory remains stable.

### 5.4 Ablations

The ablation study isolates the mechanisms that distinguish PROPRIOVLA from simple state conditioning. We evaluate four variants: (1) removing the visual self-grounding loss $\mathcal{L}_{\text{ground}}$; (2) replacing normalized KL with an unnormalized localization loss; (3) using raw optical flow instead of state-aligned residual motion; and (4) replacing the learned grounding queries with state-derived queries. Each ablation is evaluated with the same success, action, trajectory-shift, and grounding metrics.

These ablations test the central mechanism: the policy must use proprioception to mine visual self-grounding targets, rather than merely receive proprioceptive values as extra inputs. The comparison separates three alternatives: gains from additional state, gains from motion saliency alone, and gains from learned-query self-grounding.

## 6 Limitations

PROPRIOVLA assumes that the robot is at least partially visible during training. If the embodiment is consistently outside the camera view, the state-aligned motion signal cannot provide useful self-grounding supervision, and the visual query has no direct evidence from which to recover the robot's own state.

State-aligned residual motion can be ambiguous when non-robot motion is coupled to the robot. Grasped objects, deformable materials, moving backgrounds, or other agents may move synchronously with proprioceptive change and receive high alignment scores. Confidence weighting reduces weak supervision, but it cannot fully distinguish the robot body from every object moved by that body.

Finally, the evaluation covers a finite set of embodiments, camera perturbations, and manipulation tasks. Broader multi-camera setups, mobile manipulation, multi-robot scenes, long-horizon tasks, and highly dynamic environments remain important tests for whether proprioceptive self-grounding scales beyond controlled perturbation regimes.

# 7 Conclusion

This paper studies a missing-self failure mode in VLA policies: under camera-only viewpoint shifts, predicted robot trajectories can move with the image frame even when the physical robot-scene state is unchanged. The diagnostic shows that task grounding and self-grounding are distinct requirements for spatially reliable manipulation.

PROPRIOVLA addresses this issue through two connected components: State-Aligned Residual Flow (SARF) and a Proprioceptive Alignment Module (PAM). SARF mines token-level self-grounding targets from robot interaction without manual masks or external grounding datasets, and PAM trains learned grounding queries to recover these targets with normalized KL supervision. Learning to see the self is a practical step toward VLA policies whose actions are grounded in embodiment rather than in the image frame alone.

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